§4 TIPA, PSL and ISL — part 2 of 4

Reference: subsec:m2, subsec:m3

Source: paper/section4_tipa.tex, lines 877–1683 (§4.2.2 the mechanism M2, the adaptive-evaluation mechanism; §4.2.3 the mechanism M3, the response-action mechanism), at commit e6e12cd. The Japanese manuscript is authoritative. Part 2 of §4 follows 04_tipa_1.md. Part 3 follows in 04_tipa_3.md (PSL, and the description of purpose, action and result, lines 1684–2349); part 4 (ISL, integrated high-speed processing and Automaticity, the functional axis, and the closing summary, lines 2350–2964) is in 04_tipa_4.md.

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§4.2.2 M2: the adaptive-evaluation mechanism

Reference: subsec:m2

M2 receives the present picture and the time features formed by M1, evaluates at the mechanism level, by proxy, the differences related to feasibility threats, structural vulnerability and latent instability, and makes the result available to the response formation of M3.

The paper requires M2 neither to access, to hold internally, nor to compute exactly the true values of the theoretical Π, K, A_t, $E_{\mathrm{req}}(x_t)$, $\mathrm{Crit}^{(1)}(x_t)$, $\mathcal{D}_{\mathrm{FR}}$, $\mu_{\mathrm{eff},t}$, the robust persistence horizon Hor_f, Q and others. The evaluation of M2 depends on information available from M1 and others (rem:theory_quantity_vs_internal of §3).

Definition (def:m2_eval_space): internal-evaluation information space of M2 (classification B). The type of internal evaluation information that M2 provides to M3 is a non-empty set $\mathfrak{E}$, with $\mathrm{ev}_t \in \mathfrak{E}$. The internal representation form of $\mathfrak{E}$ is not fixed uniquely in Phase 1.

Definition (def:request_signal_types): the list of requests and the type of request signals (classification B). A non-empty set Req of request labels and a non-empty set Qual of qualities of a request are placed. By a fixed readout $\rho_{\mathrm{req}} : \mathcal{M} \to \mathcal{P}_{\mathrm{fin}}(\mathsf{Req})$, the *list of requests* at time t is $\mathrm{Req}_t := \rho_{\mathrm{req}}(M_t)$. The space of request signals is $\mathsf{Sig} := \mathcal{P}_{\mathrm{fin}}(\mathsf{Req} \times \mathsf{Qual})$; the *request signal* output by M2 is $\mathrm{sig}_t \in \mathsf{Sig}$, and in the standard form $\mathrm{sig}_t \subseteq \mathrm{Req}_t \times \mathsf{Qual}$. $(q, c) \in \mathrm{sig}_t$ means that the request q is raised at time t as a request of quality c. The enumeration and abstract classification of the elements of Qual are not given in Phase 1 and are a task of Phase 1.α. The symbol Qual is unrelated to the relation quality $Q_{i \leftarrow j,t}$ of §6. The request signal sig_t is a quantity distinct from the three-principle evaluation vector s_t of §3 (def:S3).

Definition (def:rq_eval_types): the type of the request-by-object evaluation (classification B). A non-empty set $\mathfrak{E}^{\mathrm{rq}}$ of evaluation values is placed, and $\mathfrak{E}^{\mathrm{rq}}_{\mathrm{map}} := \bigcup_D (\mathfrak{E}^{\mathrm{rq}})^D$, where D ranges over finite subsets of $\mathsf{Req} \times (\mathsf{Lab} \cup \{*\})$. The *request-by-object evaluation* output by M2 is $\mathrm{ev}^{\mathrm{rq}}_t \in \mathfrak{E}^{\mathrm{rq}}_{\mathrm{map}}$ with domain D^{rq}_t; in the standard form $D^{\mathrm{rq}}_t \subseteq \mathrm{Req}_t \times (\mathcal{K}_t \cup \{*\})$. $(q, k) \in D^{\mathrm{rq}}_t$ means that token k was evaluated in the light of request q, and $(q, *)$ that the evaluation was made without going through an object. A pair not in the domain is *unevaluated* and is not treated as a low evaluation value.

Definition (def:m2_map): adaptive-evaluation mechanism M_2 (classification B). M2 is a deterministic standard map

$$\mathrm{M2} : X \times E_{\mathrm{ext}} \times \Xi \times \Delta^{n-1} \times \mathcal{D}_{M1} \times \mathcal{M} \times X^{\leq W} \times \mathcal{J}^{\leq W_{\mathrm{short}}} \to \mathcal{P}_{\mathrm{prio}} \times \mathfrak{E} \times \mathsf{Sig} \times \mathfrak{E}^{\mathrm{rq}}_{\mathrm{map}},$$

with $(P_t, \mathrm{ev}_t, \mathrm{sig}_t, \mathrm{ev}^{\mathrm{rq}}_t) = \mathrm{M2}(\hat{x}_t, \hat{e}_t, \xi_t, w_t, D_{M1,t}, M_t, \{\hat{x}_s\}, \{I_{\mathrm{sense},s}\})$. M2 does not take as input the outputs of M3 or the evaluation of M3 by the purpose criterion (def:eval_vs_judgment), neither at the same time nor through retained records. In the standard form the value of M2 does not depend on the purpose-objecthood components $\xi_{\mathrm{tgt},t}$ and $\xi_{\mathrm{tgt},t}(k)$ of the discrimination state, nor on the part of the components of M_t that carries retained records of M3 outputs and of evaluations by the purpose criterion. The time features $D_{M1,t}$ are thus an input of M2, consistent with the generation dependence $M_1 \to M_2$.

Scope of the non-return to M2 (`rem:m2_no_return_scope`). The condition of `def:m2_map` forbids taking as inputs of M2 the outputs of M3, the evaluation by the purpose criterion, the judgments and their retained records, and using the purpose-objecthood components of the discrimination state. It does not mean that information originating in M3 cannot affect M2 at later times. The paper does not exclude two indirect influences:

- (a) An effective action changes the state and the external environment state, and the result appears, through sensing and the integration of M1, in the input of M2 at a later time. Actions that intentionally change the internal cognitive state, $u_t \in U_{\text{int}}$ (`rem:action_space_structure`), are included.
- (b) M1 uses the retained content M_t (including the record of the set of purposes) to allocate attention or perform other integration, and its output becomes the input of M2. The map of M1 takes M_t as input (`def:m1_map`), and Update_2 takes $\mathcal{M}_1$ as an argument (`def:update_operator`).

The evaluation of M2 is made on the material so given, by the persistence criterion, not by a purpose. The distinction between the evaluation and judgment of M3 by the purpose criterion and the evaluation of M2 is a distinction of the criterion of evaluation, not a causal isolation of information. The record of the final effective action u_{t-1} is distinct from the record of an M3 action command (`def:m1_map`) and is not a record of an output of M3. Whether the influences (a) and (b) actually arise depends on the realization of f, Update, sensing and M1, and this paper does not state it as a general consequence.

M2-a: detection of feasibility threats

M2-a is a classification-B standard function that evaluates, from the mechanism-level information available, an acute feasibility threat at the present time or in the near future.

Definition (`def:m2a_threat`): feasibility-threat detection function of M2-a (classification B). The mechanism-level proxy evaluation of feasibility used by M2-a is an implementation-dependent deterministic internal evaluation map $\widehat{\Pi}_{M2} : X \times \mathcal{M} \to \mathbb{R}$, with $\Pi_{\text{eval},t} := \widehat{\Pi}_{M2}(\hat{x}_t, M_t)$. $\Pi_{\text{eval},t}$ is not required to agree with the true $\Pi(x_t)$, and the ACS is not required to know the theoretical Π internally and compute $\Pi(\hat{x}_t)$ directly; it is a mechanism-level proxy under the convention of `subsec:description_levels_global`. The acute-threat judgment is $\Psi_{\text{acute}} : X \times E_{\text{ext}} \times \Xi \times \mathcal{D}_{M1} \times \mathcal{M} \to \{0,1\}$, with $C_{1,t} := \Psi_{\text{acute}}(\hat{x}_t, \hat{e}_t, \xi_t, D_{M1,t}, M_t)$. $C_{1,t} = 1$ means that M2 has detected an acute threat, and is not identified with the truth or falsity of the time-indexed satisfaction predicate SM.

Definition (`def:s4_internal_viability_check`): internal feasibility judgment and theoretical feasibility judgment (classification C). If an internal judgment rule is adopted, it is described by a fixed map $\Psi_K : \mathcal{M} \times \mathcal{R}_{\text{int}} \to \{0, 1, \perp\}$, where $\perp$ means undecided. Only when $\Psi_K(M_t, r_t) \in \{0,1\}$ is agreement or disagreement with $\mathbf{1}[\Pi(x_t) \geq 0]$ compared. Undecided is not counted as disagreement or as false, and this comparison is not required as an input of M2. Agreement of the sign of $\Pi_{\text{eval},t}$ with this judgment is not assumed either.

M2-b: evaluation of structural vulnerability

M2-b is a classification-B standard function that evaluates by proxy, from the viewpoints of inability to respond, lack of means, lack of resources, lack of support and degradation of relation quality, the insufficiency of the response structure against a present threat or request. The relation here means the minimal structure of ACS relations of §6, and §4 does not redefine its classification or relation descriptors. The name of the type "degradation of relation quality" refers to a type of internal evaluation of feasibility-related differences accompanying external relations; it is not a proxy value of the relation quality $Q_{i \leftarrow j,t}$ of §6, which is a quantity of the describer (`rem:Q_internal_formal`).

Definition (`def:mech_mu_eff`): general form of the mechanism-level effective action diversity (classification C). For an action set $\hat{A} \subseteq U$ estimated at the mechanism level to be available, and a deterministic equivalence relation $\hat{\sim}$ on it, $\hat{\mu}_{\text{eff}}(\hat{x}; \hat{A}, \hat{\sim}) := |\hat{A}/\hat{\sim}|$. It is a classification-C proxy quantity distinct from the true value of the response diversity $\mu_{\text{eff},t}$ of

§3 (def:mu_eff). $\mu_{\mathrm{eff},t}$ itself is a diagnostic quantity and is not used in the definition of the time-indexed satisfaction predicate of any principle.

Definition (def:m2b_vuln): structural-vulnerability evaluation function of M2-b (classification B). The standard vulnerabilities handled by M2-b are five types: (1) inability to respond, (2) lack of means, (3) lack of resources, (4) lack of support, (5) degradation of relation quality. With the vulnerability judgment map $\Psi_{\mathrm{vuln}} : X \times E_{\mathrm{ext}} \times \Xi \times \mathcal{M} \to \{0,1\}^5$, the detection state is $\mathbf{C}_{2,t} := \Psi_{\mathrm{vuln}}(\hat{x}_t, \hat{e}_t, \xi_t, M_t) \in \{0,1\}^5$. The concrete judgment rule of each component depends on the implementation.

If M2 uses a coarse estimate of action possibility, it may construct $\hat{A}_{M2,t} := \mathrm{Available}_{M2}(\hat{x}_t, \hat{e}_t, M_t)$ with $\mathrm{Available}_{M2} : X \times E_{\mathrm{ext}} \times \mathcal{M} \to \mathcal{P}_{\mathrm{fin}}(U)$. $\hat{A}_{M2,t}$ is not identified with the true feasible action set A_t. If M2 uses a predicted-response equivalence relation $\hat{\sim}_{M2,t}$, then $\hat{\mu}_{\mathrm{eff},M2,t} := \hat{\mu}_{\mathrm{eff}}(\hat{x}_t; \hat{A}_{M2,t}, \hat{\sim}_{M2,t})$. With $\mathrm{Eq}(U)$ the set of all equivalence relations on the non-empty set U, a fixed map $\Phi_{\mathrm{eq}} : \mathcal{M} \to \mathrm{Eq}(U)$ is given, and $\hat{\sim}_{M2,t}$ is the restriction of $\Phi_{\mathrm{eq}}(M_t)$ to $\hat{A}_{M2,t}$. For an empty estimated action set the cardinality of the quotient is 0. The concrete equivalence rule depends on the implementation.

M2-c: mechanism-level evaluation of latent instability

M2-c is the third classification-B standard function; it evaluates a latent instability in which a short-term improvement of the feasibility margin coexists with a reduction of the response diversity estimated at the mechanism level. The indices used as its evaluation result are distinguished as classification C.

Definition (def:m2c_latent_function): latent-instability evaluation function of M2-c (classification B). From the present and available past values of Π_{eval}, $\hat{\mu}_{\mathrm{eff},M2}$ and others, M2-c is a standard evaluation function that can detect the coexistence of short-term improvement and reduction of response diversity. The comparison window, the method of history reference and the concrete judgment rule depend on the implementation.

Definition (def:m2_latent_instability): mechanism-level index of latent instability (classification C). A comparison interval $\Delta t_{\mathrm{long}} \in \mathbb{N}$ is fixed. The map reading retained records is $\rho_{\mathrm{eval,past}} : \mathcal{M}^{\mathrm{W}} \to (\mathbb{R} \times \mathbb{N}_0) \cup \{\bot\}$, where $\bot$ means that the record usable for the comparison is missing and is not a number. Only when $t \geq \Delta t_{\mathrm{long}}$ and the readout returns a pair (a, b): $\Delta\Pi_{\mathrm{long},t} = \Pi_{\mathrm{eval},t} - a$, $\Delta\hat{\mu}_{\mathrm{eff,long},t} = \hat{\mu}_{\mathrm{eff},M2,t} - b$, and $C_{3,t} = \mathbf{1}[\Delta\Pi_{\mathrm{long},t} > 0 \wedge \Delta\hat{\mu}_{\mathrm{eff,long},t} < 0]$. a, b are retained records of the values at the time of comparison and are not necessarily the past true values. For a short history or a missing record, $C_{3,t}$ is not defined and is not replaced by false or 0. The total map of M2 also fixes a response for the missing case, but that is not the value of $C_{3,t}$. The index is not identical with T-STAB or with a time-indexed satisfaction predicate of the three principles.

Priority warnings and request signals

Definition (def:m2_priority): Priority warning system (classification B). The categories of $\mathcal{P}_{\mathrm{prio}}$ are interpreted as follows.

- **Priority 1:** a warning on acute danger or deterioration of feasibility.
- **Priority 2:** a warning on structural vulnerability in means, resources, support, relations and others.
- **Priority 3:** a warning on a sign of future vulnerabilization or degeneration of response possibility.
- **None:** the state in which none of the standard warnings above is generated.

The concrete priority rule depends on the implementation within the range that keeps the meanings above.

Definition (def:m2_request_signal): generation of request signals by M2 (classification B). M2 generates the request signal sig_t on the basis of evaluations related to feasibility and requests. sig_t does not specify the operating mode of the response. The rule of which request is raised as a request of which quality depends on the implementation. Evaluations of internal

need (for example, an internal signal of deficiency or a warning of remaining electric power) can appear as elements of sig_t (rem:m2_need_evaluation).

Reading of the request-by-object evaluation (rem:rq_evaluation_reading). When several requests are directed at the same token, the evaluation is given per request. For example, the token of the same large animal can be evaluated as prey in the light of a request for food and as a threat in the light of a request for safety. The evaluation depends on the retained content M_t and may disagree with the theoretical consequences; an individual that does not remember the animal's danger can evaluate the same object only as prey. M2 does not use the purposes or judgments of M3 in evaluation. A switch of requests is expressed across times: M1 and M2 sense and evaluate a situation produced as a result of an action (for example, danger increased because the opponent could not be overpowered), and the request signal and evaluation change at the next time. No dedicated symbol is placed on how heavily each evaluation is treated.

The conceptual generation dependence inside M2 is organized as: M2-a / M2-b / M2-c $\rightarrow$ $(P_t, \mathrm{ev}_t) \rightarrow (\mathrm{sig}_t, \mathrm{ev}_t^{\mathrm{rq}})$. This is an organization of information dependence and does not fix the order of execution (rem:tipa_bidirectional).

Information quality of the M2 evaluation and its non-quantitative character (rem:error_bounds). No universal error bound, exact agreement or convergence is required between the internal evaluation of M2 and the theoretical quantities, including the truth or falsity of the time-indexed satisfaction predicates of the three principles. The paper therefore does not state as a general consequence, from the firing of a Priority, that some time-indexed satisfaction predicate of the three principles is false or that blockage holds, nor, from the absence of firing, the absence of these.

Internal evaluation of need and critical required factors (rem:m2_need_evaluation). The acute-threat detection of M2-a and the evaluation of lack of resources and lack of support in M2-b can be standard places for internally evaluating the differences related to the critical required factors of §3 (def:eu_fire). An internal signal of deficiency in organisms and a warning of remaining electric power in artificial systems are examples of such internal evaluation. The paper does not require agreement between the factors internally evaluated as needed and the theoretical one-step critical required factor set $\mathrm{Crit}^{(1)}(x_t)$. That a factor that is theoretically critical is not internally evaluated as needed (a miss), and that a factor that is not critical is internally evaluated as needed (a false evaluation), are both not excluded by the standard functional forms of §4. Such internal evaluation can arise both in a phase where the present state is inside K (maintenance) and in one where it is outside (recovery). The truth condition of the time-indexed satisfaction predicate of EU is determined only by the theoretical $\mathrm{Crit}^{(1)}(x_t)$ and does not use the evaluation result of M2 (rem:crit_description_level). §4 places no dedicated symbol for the internal evaluation of need; its result is treated as possibly contained in $C_{1,t}$, $\mathbf{C}_{2,t}$, P_t, ev_t and the request signal sig_t.

Many-to-many correspondence of TIPA and the three principles (rem:m2_three_principles). M2-a, M2-b, M2-c and Priority 1, 2, 3 are not fixed one-to-one to any one of SM, EU, FR. Likewise M1 is not identified one-to-one with SM, nor the response formation of M3 with FR. The correspondence between principles and mechanisms is many-to-many (def:principles_tipa_connection of §3).

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§4.2.3 M3: the response-action mechanism

Reference: subsec:m3

M3 takes the information made available from M1 and M2 as material for judgment, evaluates by the purpose criterion, judges the operating mode, the purposes and the state of the purposes, constructs and compares action candidates, and forms the TIPA-side action command to be handed to PSL.

Definition (`def:eval_vs_judgment`): distinction of two evaluations and a judgment (classification B). Three are distinguished in §4:

1. *evaluation by the persistence criterion* (M2): giving values to a state, a request or an object on the basis of the ACS's feasibility related to the three principles and its fundamental requests. It arises as an expression of the structure of the ACS concerned; the outputs P_t, ev_t, sig_t, $\mathrm{ev}_t^{\mathrm{rq}}$ of M2 are this;
2. *evaluation by the purpose criterion* (M3): giving values to candidates, situations or the progress of purposes in the light of purposes, preferences and other criteria used in thinking. It is processed inside M3, and the candidate fit comparison fit_t (`def:m3_fit`) is one such evaluation;
3. *judgment* (M3): deciding by comparing the operating mode, the purposes, the state of the purposes and the action candidates, with the two evaluations and the integrated information as material. The outputs mode_t, Pur_t, stat_t, $u_{\mathrm{M3},t}$ of M3 are this.

In §4, "evaluation" without qualification means (1), and (2) is written "evaluation by the purpose criterion". M3 does not generate an evaluation by the persistence criterion. The evaluation by the purpose criterion and the judgment do not return to the input of M2 (`def:m2_map`). A tendency of choice that appears without thinking or deliberation is described not as an evaluation by the purpose criterion but as bias (`def:bias_tendency` of §5), and is also distinguished from evaluation by the persistence criterion.

Definition (`def:m3_map`): response-action mechanism M_3 (classification B). M3 is a deterministic standard map

$$\mathrm{M3} : X \times E_{\mathrm{ext}} \times \Xi \times \mathcal{P}_{\mathrm{prio}} \times \mathfrak{C} \times \mathcal{P}_{\mathrm{fin}}(\mathsf{Req}) \times \mathsf{Sig} \times \mathfrak{C}_{\mathrm{map}}^{\mathrm{rq}} \times \mathcal{P}_{\mathrm{fin}}(\mathsf{Pur}) \times \mathcal{M} \to U \times \mathsf{Mode} \times \mathsf{Style} \times \mathcal{P}_{\mathrm{fin}}(\mathsf{Pur}) \times \mathsf{St},$$

with $(u_{\mathrm{M3},t}, \mathrm{mode}_t, \mathrm{style}_t, \mathrm{Pur}_t, \mathrm{stat}_t) = \mathrm{M3}(\hat{x}_t, \hat{e}_t, \xi_t, P_t, \mathrm{ev}_t, \mathrm{Req}_t, \mathrm{sig}_t, \mathrm{ev}_t^{\mathrm{rq}}, \mathrm{Pur}_{t-1}, M_t)$. Pur_{t-1} is the retained record of the set of purposes at the previous time (`def:s4_carrier_contract`). St and stat_t are defined in `def:purpose_status`. The domain of M3 contains arbitrary elements of each type for the list of requests, the request signal and the request-by-object evaluation; the value of M3 is therefore defined also for an input from which part of the requests has been removed (`def:req_involvement`).

M3-a: constructing information for response formation

M3-a constructs the information already generated by the upstream mechanisms as integrated input information usable for candidate formation and comparison. M3-a does not regenerate Priority, the discrimination state, the request signal or the evaluations.

Definition (`def:m3a_info`): integrated input information of M3-a (classification B). The integrated input space of M3 is $\mathcal{J}_{M3} := \mathcal{P}_{\mathrm{prio}} \times \mathfrak{C} \times \mathcal{P}_{\mathrm{fin}}(\mathsf{Req}) \times \mathsf{Sig} \times \mathfrak{C}_{\mathrm{map}}^{\mathrm{rq}} \times \mathcal{P}_{\mathrm{fin}}(\mathsf{Pur}) \times X \times E_{\mathrm{ext}} \times \Xi \times \mathcal{M}$, and the integrated input at time t is $I_{M3,t} := (P_t, \mathrm{ev}_t, \mathrm{Req}_t, \mathrm{sig}_t, \mathrm{ev}_t^{\mathrm{rq}}, \mathrm{Pur}_{t-1}, \hat{x}_t, \hat{e}_t, \xi_t, M_t) \in \mathcal{J}_{M3}$.

M3-b: candidate formation and comparison

M3-b forms purposes, constructs the available action candidates, and compares internal fit differences against the purposes and requests.

Definition (`def:purpose_formation`): the type of purposes and the formation of purposes (classification B). The set of directions of a purpose is $\mathsf{Dir} := \{\mathsf{self}\} \cup \mathsf{Lab} \cup (\{\mathsf{self}\} \times \mathsf{Lab})$: self takes the self state as its direction, $k \in \mathsf{Lab}$ an internal-object token, and (self, k) the relation between the self and token k. A non-empty set Cnt of contents of a purpose is placed, and the set of purposes is $\mathsf{Pur} := \mathsf{Req} \times \mathsf{Dir} \times \mathsf{Cnt}$. For a purpose $p = (q, d, z)$, q is called the *fundamental request* of p. The finite set of purposes that M3 forms as a judgment at time t is $\mathrm{Pur}_t \in \mathcal{P}_{\mathrm{fin}}(\mathsf{Pur})$ (`def:m3_map`), and $\mathrm{Pur}_t = \emptyset$ is allowed. The material for forming purposes is limited to the inputs of M3, including the integrated information and retained records of M1 and the evaluations and request signals of M2. The request signal is one of the main sources of a purpose, but a continuing purpose remaining in retained records, and tokens and their discrimination, can also be sources; the paper does not claim that the sources are exhaustive. The enumeration

of the elements of Cnt, the rule of formation, and the order, inside M3, of formation and the judgment of the operating mode depend on the implementation.

Hierarchy and nesting of purposes (`rem:purpose_hierarchy`). A purpose is read in the order: fundamental request, proximal purpose consisting of direction and content, action, result. A purpose whose direction is self needs no target token. Several purposes with the same fundamental request, standing together at one time or across times, are read as nesting of purposes. For example, "looking for water" on the basis of thirst and "drawing" toward the token of a water place that has been found share the same fundamental request. The definition of the relation between means and purpose itself is a task of Phase 1.α (`subsec:phase1a`). Becoming a purpose is a change on the side of internal tokens; it is not read as an external component becoming an action-response component through being made a purpose (`rem:Eext_decomposition_reading`).

Direction of a purpose and internal-object tokens (`rem:purpose_direction_token`). A direction $k \in \mathrm{Lab}$ of a purpose is a reference to the label of an internal-object token and does not express temporal identity of an external individual. The same label can represent a different object across times (`rem:token_generation`), so that a purpose direction continuing to point at the same external object across times is a separate condition. The rule of token persistence is left to §5. Since a direction is defined on Lab, at a time with $k \notin \mathcal{K}_t$ (for example, when the object is no longer sensed), M3 can hold a purpose directed at k and judge its state. On the other hand, since the domain of the standard-form request-by-object evaluation is contained in $\mathrm{Req}_t \times (\mathcal{K}_t \cup \{*\})$ (`def:rq_eval_types`), M2 does not evaluate the pair (q, k) at that time and can only give an evaluation $(q, *)$ of the overall row.

Definition (`def:purpose_status`): judgment of the state of a purpose (classification B). The set of judgment values is $\{\mathsf{cont}, \mathsf{ach}, \mathsf{term}, \mathsf{und}\}$, and $\mathrm{St} := \bigcup_P \{\mathsf{cont}, \mathsf{ach}, \mathsf{term}, \mathsf{und}\}^P$, where P ranges over finite subsets of Pur. The *judgment of the state of purposes* that M3 outputs at time t is $\mathrm{stat}_t \in \mathrm{St}$, and in the standard form its domain is the record Pur_{t-1} of the set of purposes at the previous time. $\mathrm{stat}_t(p) = \mathsf{cont}, \mathsf{ach}, \mathsf{term}, \mathsf{und}$ express respectively continuing the purpose p, judging it attained, ending it without attaining, and withholding the judgment. term includes abandonment and switching to another purpose; the distinction of reasons is a task of Phase 1.α. In the standard form, if $\mathrm{stat}_t(p) = \mathsf{cont}$ then $p \in \mathrm{Pur}_t$, and if $\mathrm{stat}_t(p) \in \{\mathsf{ach}, \mathsf{term}\}$ then $p \notin \mathrm{Pur}_t$. When $\mathrm{stat}_t(p) = \mathsf{und}$, whether p remains in Pur_t is not determined.

Reading of the judgment of the state of a purpose (`rem:purpose_status_reading`). Attainment of a purpose is determined by M3 as a judgment. A change in the evaluation of M2 (for example, a lowered danger evaluation or the release of a warning) is material for judgment and is not itself read as a judgment of attainment. M3 judges, with the evaluation of M2, retained content, integrated information of M1 and predictions as material, through evaluation by the purpose criterion in the light of the purpose. If it judges that confirmation is insufficient, it sets cont or und and can include an information-acquiring action among the candidates. The amount of information used for the judgment, the detail of comparison and whether additional confirmation is made can depend on the operating mode and the implementation; the paper does not fix the procedure. In a simple ACS a purpose may cease to be formed because the request disappears, without an explicit judgment of attainment. This is described as the purpose not remaining in Pur_t under und, and the paper does not require an explicit judgment of attainment of every ACS. The internal judgment need not agree with theoretical attainment (`def:purpose_course`).

Definition (`def:m3_candidate_set`): the candidate set of M3 (classification B). The candidate set that M3 handles at time t for response formation is $\mathrm{Cand}_t \subseteq U$, $\mathrm{Cand}_t \neq \emptyset$. Even when no other ordinary candidate exists, using the selective waiting $u_\emptyset \in U$ of §2, $\mathrm{Cand}_t = \{u_\emptyset\}$ can occur. Cand_t is not identified with the true feasible action set A_t, and no inclusion in either direction is assumed in general. The consequence of the final effective action satisfying $u_t \notin A_t$ follows `prop:selection_responsibility` of §3 ($x_{t+1} \notin K$). The paper places $u_t \in A_t$ neither as a constitutive requirement nor as an assumption (`rem:selection_responsibility`).

Definition (`def:m3_fit`): candidate fit comparison (classification B). The map describing the internal fit difference among candidates is $\mathrm{fit}_t : \mathrm{Cand}_t \to \mathbb{R}$. fit_t may depend on M2 evaluation information, the internal model, state and environment estimates, Priority, retained information and others. It is not identified with the true Π, A_t, $\mathrm{Crit}^{(1)}(x_t)$, $\mu_{\mathrm{eff},t}$, Q or others. fit_t is one evaluation by the purpose criterion (`def:eval_vs_judgment`) and is not an evaluation

by the persistence criterion.

Definition (`def:s4_candidate_sources`): provenance of candidates and fit comparison (classification B). Fixed $\Phi_{\mathrm{cand}} : \mathcal{J}_{M3} \to \mathcal{P}(U) \setminus \{\emptyset\}$ and $\Phi_{\mathrm{fit}} : \mathcal{J}_{M3} \times U \to \mathbb{R}$ give $\mathrm{Cand}_t = \Phi_{\mathrm{cand}}(I_{M3,t})$ and $\mathrm{fit}_t(u) = \Phi_{\mathrm{fit}}(I_{M3,t}, u)$ for $u \in \mathrm{Cand}_t$. Compactness of candidates and attainment of a maximum are not assumed; action determination follows the already defined deterministic selection rule.

Use of per-candidate predictions (`rem:candidate_prediction`). M3 may use, as material for comparison, $\hat{G}_t(u)$, the prediction map $\hat{g}_t$ of the internal model (`def:env_set_prediction`) applied to a candidate $u \in \mathrm{Cand}_t$. This is an application of a prediction, not the generation of an evaluation by the persistence criterion; an evaluation by the purpose criterion can be given to the result of the application. The prediction set for the final effective action is recorded after u_t is determined (`def:prediction_record_check`). No description is placed that uses a prediction about u_t in the same-time comparison.

Fit to requests versus discrimination (`rem:fit_vs_identity`). M1-b discriminates objects or information internally, and M3-b compares the difference in fit of action candidates to the present purposes and requests. The paper therefore does not state, as a general consequence, the fit of a particular action candidate from the discrimination result of M1 alone.

When information is insufficient, in an implementation in which information acquisition itself is defined as an action in U, an information-acquiring action may be included in Cand_t.

M3-c: judgment of the operating mode and determination of the action command

M3-c determines the one TIPA action command to be handed to PSL, on the basis of the judgment of the operating mode and the candidate comparison of M3-b.

Definition (`def:mode_judgment`): judgment of the operating mode (classification B). $\mathrm{mode}_t \in \mathsf{Mode}$ is an output of M3 (`def:m3_map`), judged by M3 with the request signal sig_t, Priority P_t, the evaluations and others as material. M2 does not specify an operating mode. The procedure of judgment, and the concrete candidate-comparison procedure inside each mode, depend on the implementation.

Definition (`def:request_response_mismatch`): mismatch between request and response (classification C). A relation $\mathsf{Corr} \subseteq \mathsf{Qual} \times \mathsf{Mode}$ given by the implementation and fixed expresses the correspondence between the quality of a request and an operating mode. For a request $(q, c) \in \mathrm{sig}_t$ at time t with $(c, \mathrm{mode}_t) \notin \mathsf{Corr}$, the request is said to be *mismatched* with the response at time t. The set of mismatched requests is $\mathrm{Mis}_t := \{(q, c) \in \mathrm{sig}_t \mid (c, \mathrm{mode}_t) \notin \mathsf{Corr}\}$. The content of Corr is not given in Phase 1. The paper does not state as a general consequence that a mismatch originates in a defect of M2 or M3, that an unfavorable result arises from a mismatch, or the converses of these.

Definition (`def:M3_modes`): operating modes of M3 (classification B). The modes of Mode are interpreted as follows.

- **3A, emergency-response mode:** a standard mode that, in response to an evaluation or request showing high urgency, gives relative priority to speed of response formation over detailed comparison.
- **3B, deliberative-response mode:** a standard mode that compares several candidates and several evaluation differences in relatively fine detail.
- **3C, ordinary-response mode:** a standard mode for when neither urgency nor a particular demand for detailed comparison is dominant.

Each mode is not a label expressing the truth or falsity of a time-indexed satisfaction predicate of the three principles.

Distinction of M3-3A and PSL-4 (`rem:m3_3a_vs_psl4`). 3A is an operating mode in which emergency response is formed within TIPA on the basis of information available to M1, M2, M3, and differs from PSL-4, a local protective response that does not generally require completion of detailed TIPA processing. The paper therefore does not state as a general consequence the operation of PSL-4 from the holding of 3A, nor the holding of 3A from the operation of PSL-4.

Definition (`def:m3_3b_styles`): response styles of 3B (classification B). The standard response styles available within 3B are $\mathrm{Style}_{3B} := \{\text{3B-own}, \text{3B-rel}, \text{3B-comb}\}$. With $\mathrm{Style} = \mathrm{Style}_{3B} \cup \{\text{none}\}$ defined at the head of §4.2, $\mathrm{style}_t = \text{none}$ when $\mathrm{mode}_t \neq \text{3B}$. Only when $\mathrm{mode}_t = \text{3B}$ may M3 use an element of Style_{3B} as a response style as needed. The element none expresses not distinguishing a response style and does not mean the absence of response formation itself. 3B-own is a style that relatively emphasizes evaluation differences related to self-side feasibility and internal maintenance; 3B-rel a style that relatively emphasizes evaluation differences related to external relations, connection conditions and feasibility-related differences accompanying external relations; 3B-comb a style that integrates several evaluation differences by an implementation-dependent rule. These do not mean a ranking of normative good and bad or of ethical superiority.

Definition (`def:action_decision`): determination of the TIPA action command (classification B). M3-c uniquely determines, by a deterministic selection rule using Cand_t, fit_t, mode_t and the response style where needed, $u_{\mathrm{M3},t} \in \mathrm{Cand}_t \subseteq U$. The comparison algorithm for each mode, the search amount, the stopping condition and the tie-break rule depend on the implementation within the range that keeps determinism and the type consistency above. $u_{\mathrm{M3},t}$ is the TIPA-side action command that M3 generates and is not the final effective action u_t entering the state transition. The relation between the two is defined in the effective-action resolution of `subsec:psl`.

M3-a, M3-b, M3-c are an organization of the roles of M3; their internal order and mutual exchange are not fixed (`rem:tipa_bidirectional`). The operating mode, the purposes and the action command are all determined as outputs of M3 of `def:m3_map`.

No guarantee of success of M3 (`rem:m3_output_quality`). This paper does not state as a general consequence, from M3 having determined $u_{\mathrm{M3},t}$, that $u_{\mathrm{M3},t} \in A_t$, the truth of a time-indexed satisfaction predicate of the three principles, avoidance of blockage, avoidance of collapse, or attainment of an expected result.

Check status. This summary adds no new definition, assumption or result. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.